

VEHICLE-TO-GRID (V2G) AND GRID-TO-VEHICLE (G2V)

Abstract

The increasing adoption of electric vehicles (EVs) has introduced new possibilities for efficient energy interaction between transportation systems and power grids. Grid-to-Vehicle (G2V) technology allows electric vehicles to charge their batteries using electrical energy supplied by the power grid in a controlled and efficient manner. This process helps manage charging demand and ensures reliable energy utilization. Vehicle-to-Grid (V2G) technology enables bidirectional power flow, allowing electric vehicles to deliver stored electrical energy back to the grid during peak demand or emergency situations. By supporting load balancing, frequency regulation, and renewable energy integration, V2G improves grid stability and reliability. Together, V2G and G2V technologies form an intelligent energy management system that enhances smart grid performance, reduces energy costs, and promotes sustainable transportation. These technologies play a vital role in the future of clean energy systems and electric mobility.